

# Document for

Beyond Script Family Boundaries: Towards Unified Open-Set Scene Text Recognition

## Requirement / Our spec

CPU: anything with AVX2 / R5 5500 (this report) and 12500 (dev)

\* Intel or AMD, I have no idea if this works on Zhaoxin or smth to that nature

GPU (inference): 3GiB Nvidia GPU (Pascal or newer) / GTX1070 (this report)

GPU (training): 48 GiB Nvidia GPU / RTX A6000 (local) and A40 (Alvis)

\*\* early prototypes are trained on A100-40 GPUs on Berzelius, it may still work

Ram (inference): 8 GiB / 8 GiB

Ram (training) : 64GiB / 70 Gib (local) and 64GiB (Alvis)

Disk: 500GiB / 500GiB (for this report, darn used disks in EU are EXPENSIVE)

OS: linux / Debian 12.14 (for this report) and Archlinux (for dev)

Internet: Yes / Yes

Note: this document is not for power users.

This is **NOT** how things should be done,

but rather a quick way to just run it on a testbench.

Audit your code and script and set environments properly,

**and please do not blindly run random bash scripts off the internet.**

Trivia: Maunl, aka Pallas' cat, is the oldest cat species on earth today.

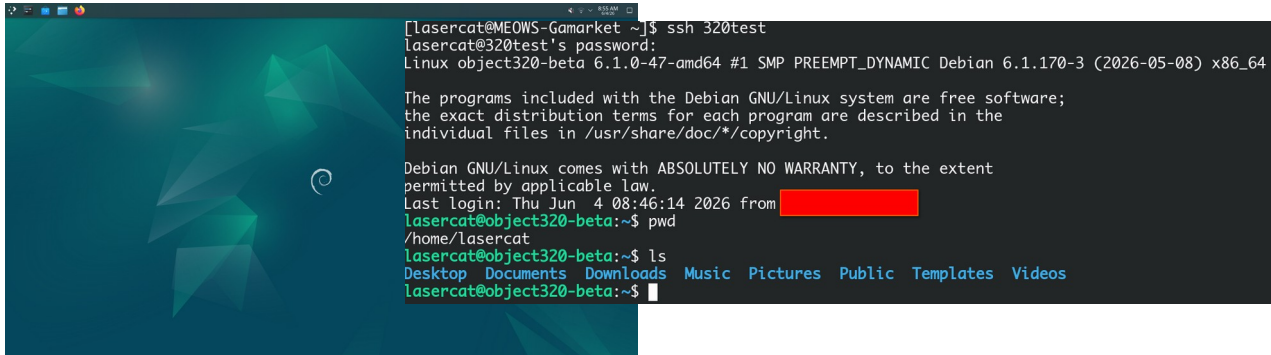


# 1. Setup Environment

## System & Cuda

1. install debian 12.14 KDE, select ssh if you need it, use lasercat as username

you can mod the code later to fit your username and paths...



```
[lasercat@MEOWS-Gamarket ~]$ ssh 320test
lasercat@320test's password:
Linux object320-beta 6.1.0-47-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.170-3 (2026-05-08) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Jun  4 08:46:14 2026 from [REDACTED]
lasercat@object320-beta:~$ pwd
/home/lasercat
lasercat@object320-beta:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
lasercat@object320-beta:~$
```

2. download script from <https://github.com/lancercat/320-env-debian> to ~/Downloads

3. set up basic environment

```
su
```

```
bash basicenv.sh
```

4. **reboot** (bcs the kernel version may bump, so this is a **MUST**)

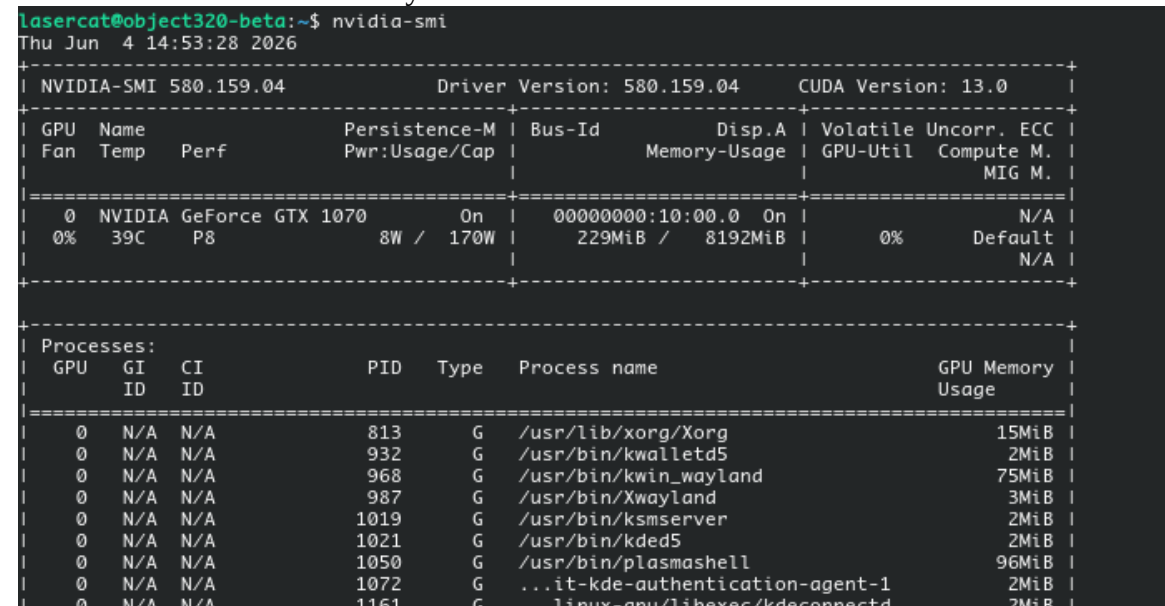
5. install cuda

```
su
```

```
bash setupenv.sh
```

6. reboot (for nvidia driver to kick in)

7. run nvidia-smi to check that you have it



```
lasercat@object320-beta:~$ nvidia-smi
Thu Jun  4 14:53:28 2026

+-----+
| NVIDIA-SMI 580.159.04                  Driver Version: 580.159.04          CUDA Version: 13.0                  |
+-----+-----+
| GPU   Name                               Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp  Perf              Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Compute M. |
|=====+=====+
| 0  NVIDIA GeForce GTX 1070                On | 00000000:10:00.0 On |         N/A |
| 0%   39C   P8              8W / 170W | 229MiB / 8192MiB |      0%   Default |
|=====+=====+
+-----+-----+

+-----+
| Processes:                               GPU Memory                               |
|  GPU   GI    CI          PID    Type   Process name                          Usage  |
|=====+=====+
|  0     N/A  N/A         813     G   /usr/lib/xorg/Xorg                     15MiB |
|  0     N/A  N/A         932     G   /usr/bin/kwalletd5                     2MiB  |
|  0     N/A  N/A         968     G   /usr/bin/kwin_wayland                  75MiB |
|  0     N/A  N/A         987     G   /usr/bin/Xwayland                      3MiB  |
|  0     N/A  N/A        1019     G   /usr/bin/ksmsserver                    2MiB  |
|  0     N/A  N/A        1021     G   /usr/bin/kded5                         2MiB  |
|  0     N/A  N/A        1050     G   /usr/bin/plasmashell                   96MiB |
|  0     N/A  N/A        1072     G   ...it-kde-authentication-agent-1       2MiB  |
|  0     N/A  N/A        1161     G   ...linux-gnu/libexec/kdeconnectd      2MiB  |
+-----+-----+
```

## Torch & Etc

1. install torch and torch\_scatter (a custom fork to fix glog)

```
source ~/.bashrc && bash pyenv.sh
```

2. set up code, weight, and models.

- a. prepare dir:

```
sh makdirs.sh
```

- b. clone code

```
cd ~/cat; git clone git@github.com:lancercat/uni-zero.git
```

- c. download data and weight to Downloads, put weights to ~/hydra\_saves and data to ~/ssddata

- d. link save path to /run/media/lasercat/320-eccv, or mount a disposable device there.

```
mkdir -p /run/media/lasercat/320-eccv/
```

```
mkdir -p /home/lasercat/results/
```

```
chown lasercat /home/lasercat/results/
```

```
ln -sf /home/lasercat/hydra_saves/hydra_saves /run/media/lasercat/320-eccv
```

```
ln -sf /home/lasercat/results /run/media/lasercat/320-eccv
```

```
chown lasercat /run/media/lasercat/320-eccv
```

- e. change max worker count to 1 if you are short on ram

```
vim neko_2025_NGNW/common/object_32x_presets/base_loader_factory.py
```

```
267 ][:1][:os.cpu_count()]; # to not oom alvis..
```

well if you have unlimited ram works you can just remove the piggy ([:8]) all together lol.

- ### 3. Stop and check that you have everything

- a. models & results dir

results dir should be here for the script to write into later

```
lasecat@object320-beta:~/ssddata$ ls /run/media/lasecat/320-eccv/
hydra_saves  results
```

Models should be linked here as well, you need these models.

```

lasercat@object320-beta:~/ssddatas$ ls /run/media/lasercat/320-ecvv/hydra_saves
b1jNF-1024-ostr-cntrlgnlgn-lcam-lpos-32-128-lscstSHSTT-lscstFRC_RealNI64-run2
b1dNF-1024-ostr-cntrlgnlgn-sqz-lctc-32-128-run2
b1h1NF-1024-indc2-cntrlgnl-lcam-ctc-32-128
b1h1NF-1024-indc2-cntrlgnl-lcam-ctc-32-128-run2
b1h1NF-1024-indc2-cntrlgnl-lcam-lposReal-32-128
b1h1NF-1024-indc2-cntrlgnl-lcam-lposReal-32-128-run2
b1h1NF-1024-indc2-cntrlgnl-sqz-lctc-32-128
b1h1NF-1024-indc2-cntrlgnl-sqz-lctc-32-128-run2
b1h2NF-1024-indc2-cntrlgnl-lcam-ctc-32-128-lscstSASTT
b1h2NF-1024-indc2-cntrlgnl-lcam-ctc-32-128-lscstSASTT-run2
b1h4NF-1024-indc2-cntrlgnl-lcam-ctc-32-128
b1h5NF-1024-all-cntrlgnl-lcam-ctc-32-128-SASTT
b1h5NF-1024-all-cntrlgnl-lcam-ctc-32-128-SASTT-run2
b1iNF-1024-ostr-cntrlgnl-lcam-lpos-32-128-lscstFRC_RealNI64
b1iNF-1024-ostr-cntrlgnl-lcam-lpos-32-128-lscstFRC_RealNI64-run2
b1jNF-1024-indc2-cntrlgnl-lcam-ctc-32-128-lscstFRC_RealNI64
b1jNF-1024-indc2-cntrlgnl-lcam-ctc-32-128-lscstFRC_RealNI64-run2
b1jNF-1024-indc2-cntrlgnl-lcam-ctc-32-128-lscstSASTT-lscstFRC_RealNI64
b1jNF-1024-indc2-cntrlgnl-lcam-ctc-32-128-lscstSASTT-lscstFRC_RealNI64-run2
b1jNF-1024-ostr-cntrlgnl-lcam-lpos-32-128-lscstSHSTT-lscstFRC_RealNI64
lasercat@object320-beta:~/ssddatas$
b1jNF-1024-ostr-cntrlgnl-lcam-lpos-32-128-lscstSHSTT-lscstFRC_RealNI64-run2
b1i1NF-1024-hindi-cntrlgnl-lcam-ctc-32-128
b1i1NF-1024-kannada-cntrlgnl-lcam-ctc-32-128
b1i1NF-1024-malayalam-cntrlgnl-lcam-ctc-32-128
b1i1NF-1024-marathi-cntrlgnl-lcam-ctc-32-128
b1i1NF-1024-odia-cntrlgnl-lcam-ctc-32-128
b1i1NF-1024-punjabi-cntrlgnl-lcam-ctc-32-128
b1i1NF-1024-tamil-cntrlgnl-lcam-ctc-32-128
b1i1NF-1024-telugu-cntrlgnl-lcam-ctc-32-128
b1m1NF-1024-hustobc-lscstfrc-abiaug-48-48
b1m3NF-1024-hustobc2-lscstfrc-abiaug-48-48-pe
b1NF-1024-ostr-cntrlgnl-lcam-lpos-32-128
b1NF-1024-ostr-cntrlgnl-lcam-lpos-32-128-lscstSHSTT
b1NF-1024-ostr-cntrlgnl-lcam-lpos-32-128-lscstSHSTT-run2
b1NF-1024-ostr-cntrlgnl-lcam-lpos-32-128-run2
b1NFXB4-1024-ostr-cntrlgnl-lcam-lpos-32-128
b1NFXB4-1024-ostr-cntrlgnl-lcam-lpos-32-128-run2
b2NF-1024-ostr-cntrlgnl-lcam-lposReal-32-128
b2NF-1024-ostr-cntrlgnl-lcam-lposReal-32-128-run2
mono2XL-v6S-AAF-ohem01E

```

c. check the dataset and dicts are in ~/ssddata

```
lasercat@object320-beta:~/ssddata$ ls
artdb_seen      dicts_v3        IC15_2077      lsvtdb_seen    mltrtrjp_hori  ST-abi
artdb_seen_NG   ecr-v3R        IIIT5k_3000    lsvtdb_seen_NG mltrtrjp_hv    SVT
ctwdb_seen      hustobc         istr           MJ-abi         mltrtrkr_hori  SVTP
ctwdb_seen_NG   IC03_867       loader_index   mltrtrchlat_seen rctwtrdb_seen  synthyi_5k
CUTE80          IC13_1015      lsct           mltrtrchlat_seen_NG rctwtrdb_seen_NG
```

If everything is here, you are good to go.



## 2. Reproducing the results

1. limit your card if you have a shoddy psu, its going to be a long day (or a few long days) for the card :)

su

nvidia-smi -pm 1 && nvidia-smi -pl 100

2. exit su with ctrl+D, and then activate your torch

source ~/Downloads/env/pytorch\_env/bin/activate

3. go to eval script dir

cd ~/cat/uni-zero/neko\_2025\_NGNW/beta320

trivia: uni-zero is still a beta branch of object 32x, expect breaking api changes in the near future

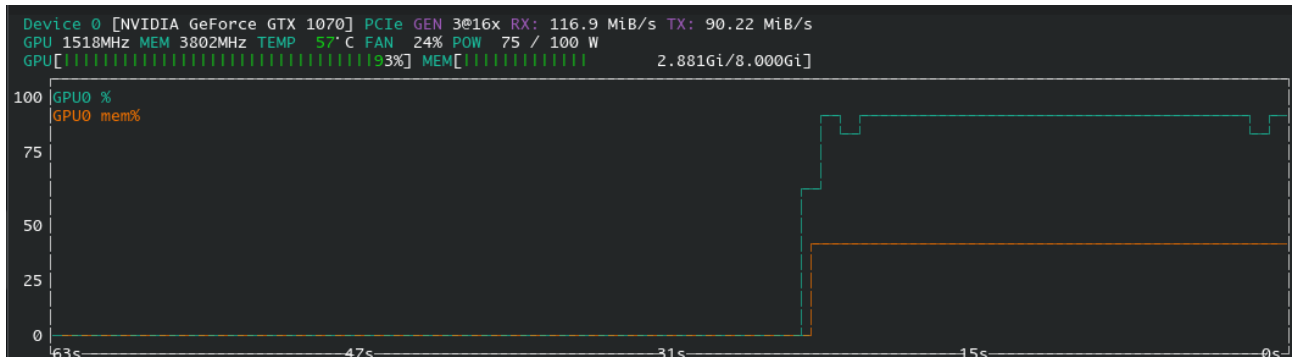
4. run main inference

PYTHONPATH=\${PWD}/../../../../ python eval\_batch.py | tee hee.log0

if you see this, its running,

```
WARN: Module notbg_oszsl-similarity cannot load OPTIMIZER_SCHED itr /run/media/lasercat/320-eccv/hydra_saves/b1h1NF-1024-indc2-cntr
align-lcam-ctc-32-128/notbg_oszsl-similarity_E1_I40000_opt_sched.pth , starting fresh
INFO: let me remind you that queue contents can not be restored for now
INFO: Module notbg_oszsl-per_inst_length_loss dose not support saving
Date: 2026-07-16-14-11-41 ,TEST: istrval_punjabi ,Epoch: 0 ,Iter: 100000 ,Total-K: 2500.0 ,ACR: 0.8216 ,FPS: 210.76252111994924
```

utilization btw, note we are running off hdd so the io is not ideal.



There is going to be a hellot results coming, go grab some cat mint and wait patiently.

5. extract extra protocol results

PYTHONPATH=\${PWD}/../../../../ python eval\_batch\_xtra.py | tee hee.log1

You will see a lot of skipping message, which is normal bcs the inference has been done already,

what we are doing here is simply extract the cached result into latex files so we can put them down

```
(pytorch_env) lasercat@object320-beta:~/cat/uni-zero/neko_2025_NGNW/beta320$ PYTHONPATH=${PWD}/../../../../ python eval_batch_extra.py | tee hee.log1
python: can't open file '/home/lasercat/cat/uni-zero/neko_2025_NGNW/beta320/eval_batch_extra.py': [Errno 2] No such file or directory
(pytorch_env) lasercat@object320-beta:~/cat/uni-zero/neko_2025_NGNW/beta320$ PYTHONPATH=${PWD}/../../../../ python eval_batch_xtra.py | tee hee.log1
/home/lasercat/cat/uni-zero/neko_2025_NGNW/beta320/../../../../
0%|
wn PC?
Skipping: /run/media/lasercat/320-eccv/results/hydra_results_eccv/b1h1NF-1024-indc2-cntralign-lcam-ctc-32-128.pt already exists.
50%| | 1/2 [00:02<00:02, 2.77s/it]
```

Trivia: you can stop and restart the inference at any time, it will skip what is done.



## 6. Get ~~DLC~~ rebuttal extra experiment results

```
cd ~/cat/uni-zero/neko_2025_NGNW/beta320-rebuttal-extra
```

```
PYTHONPATH=${PWD}/../../../../ python eval_batch_xtra.py | tee hee.log2
```

## 7. check difference with shipped paper tex file

```
cd ~/cat/uni-zero/misc
```

```
python diff_reimpl.py
```

This by default gives if there are any differences greater than 0.2%,

```
(pytorch_env) lasercat@object320-beta:~/cat/uni-zero/misc$ python diff_reimpl.py
{'summary': {}, 'raw_runs': {}}
{'summary': {}, 'raw_runs': {}}
{'summary': {}, 'raw_runs': {}}
{'summary': {}, 'raw_runs': {}}
{'summary': {}, 'raw_runs': {}}
{'summary': {}, 'raw_runs': {}}
{'summary': {}, 'raw_runs': {}}
{'summary': {}, 'raw_runs': {}}
{'summary': {}, 'raw_runs': {}}
(pytorch_env) lasercat@object320-beta:~/cat/uni-zero/misc$
```

you can adjust the script to see minor differences, say anything greater than 0.05%

```
103 try:
104     diff_report=diff_reimpl(orig,reimpl)
105     # 2. Filter for significant performance shifts (e.g., delta >= 1.0)
106     significant_changes = thresh_diff(diff_report, 0.05)
107     print(significant_changes);
108 except:
```

## Results

```
(pytorch_env) lasercat@object320-beta:~/cat/uni-zero/misc$ vim diff_reimpl.py
(pytorch_env) lasercat@object320-beta:~/cat/uni-zero/misc$ python diff_reimpl.py
{'summary': {'CamCtcCVODAcRXXII': {'Value': 0.08}, 'CamCtcCVMALAcRXXII': {'Value': 0.08}, 'CamCtcCVMALAcR': {'Std': 0.05, 'Avg': 0.01}}, 'raw_runs': {'CamCtcCVMALAcR': {'XRI': 0.0, 'XIIRI': 0.0, 'XIVRI': 0.08}, 'CamCtcCVTAACR': {'XRI': 0.0, 'XIIRI': 0.08, 'XIVRI': 0.04}, 'CamCtcCVKAAACR': {'XRI': 0.04, 'XIIRI': 0.08, 'XIVRI': 0.0}, 'SgzCtcCVMALAcR': {'XRI': 0.08, 'XIIRI': 0.0, 'XIVRI': 0.0}}, 'summary': {'BigBucCVMALAcR': {'Std': 0.04, 'Avg': 0.06}}, 'raw_runs': {'ConflCamCtcCVTAACR': {'XRI': 0.0, 'XIIRI': 0.08, 'XIVRI': 0.04}, 'BigBucCVMALAcR': {'XRI': 0.08, 'XIIRI': 0.08, 'XIVRI': 0.0}, 'ConflCamCtcCVMALAcR': {'XRI': 0.0, 'XIIRI': 0.0, 'XIVRI': 0.08}, 'ConflCamCtcCVKAAACR': {'XRI': 0.04, 'XIIRI': 0.08, 'XIVRI': 0.0}}, 'summary': {'OdiCTODAcR': {'Std': 0.0, 'Avg': 0.06}}, 'raw_runs': {'OdiCTODAcR': {'XIIRI': 0.06}}, 'summary': {'FRCCVMALAcRXXII': {'Value': 0.08}, 'ShufSACVTEAcRXXII': {'Value': 0.08}}, 'raw_runs': {'ShuffFRCCVMAACR': {'XRI': 0.08, 'XIIRI': 0.0, 'XIVRI': 0.0}, 'ShuffFRCCVTEAcR': {'XRI': 0.0, 'XIIRI': 0.0, 'XIVRI': 0.08}, 'ShuffFRCCCTHIAACR': {'XRI': 0.02, 'XIIRI': 0.06, 'XIVRI': 0.0}}, 'summary': {'UniCVHIAcRXXII': {'Value': 0.08}, 'UniCVTEAcRXXII': {'Value': 0.08}, 'UniGZSLVBEAcRXIVRII': {'Value': 0.08}}, 'raw_runs': {'UniCTMALAcR': {'XRI': 0.06, 'XIIRI': 0.02, 'XIVRI': 0.0}}, 'summary': {'SazCtcOSTRJPACRXXII': {'Value': 0.06}, 'CamCtcOSTRJPFScoreXXII': {'Value': 0.06}, 'CamCtcOSTRJPRecallXXII': {'Value': 0.1}, 'SazCtcOSTRJPJPrecisionXXII': {'Value': 0.05}}, 'raw_runs': {'SazCtcOSTRJPACR': {'XRI': 0.0, 'XIIRI': 0.05, 'XIVRI': 0.05}, 'CamLposOSTRJPACR': {'XRI': 0.0, 'XIIRI': 0.0, 'XIVRI': 0.06}, 'CamLposOSTRJPRecall': {'XRI': 0.0, 'XIIRI': 0.0, 'XIVRI': 0.05}, 'SazCtcOSTRJPRecall': {'XRI': 0.0, 'XIIRI': 0.05, 'XIVRI': 0.0}, 'CamCtcOSTRJPACR': {'XRI': 0.05, 'XIIRI': 0.0, 'XIVRI': 0.0}}, 'summary': {'ShuffFRCCOSTRJPACRXXII': {'Value': 0.05}, 'FRCCOSTRJPACRXXII': {'Value': 0.05}, 'ShuffFRCCOSTRJPRecallXXII': {'Value': 0.05}, 'XbatchOSTRJPRecallXXII': {'Value': 0.05}, 'XbatchOSTRJPRecallXXII': {'Value': 0.05}}, 'raw_runs': {'FRCCOSTRJPRecall': {'XRI': 0.0, 'XIIRI': 0.0, 'XIVRI': 0.05}, 'XbatchOSTRJPRecall': {'XRI': 0.05, 'XIIRI': 0.0, 'XIVRI': 0.0}, 'XbatchOSTRJPACR': {'XRI': 0.05, 'XIIRI': 0.0, 'XIVRI': 0.0}, 'ShufSAOSTRJPRecall': {'XRI': 0.05, 'XIIRI': 0.0, 'XIVRI': 0.0}, 'XbatchOSTRJPJPrecision': {'XRI': 0.05, 'XIIRI': 0.0, 'XIVRI': 0.0}, 'FRCCOSTRJPJPrecision': {'XRI': 0.0, 'XIIRI': 0.0, 'XIVRI': 0.05}, 'ShuffFRCCOSTRJPRecall': {'XRI': 0.0, 'XIIRI': 0.05, 'XIVRI': 0.0}, 'XbatchOSTRJPFScore': {'XRI': 0.05, 'XIIRI': 0.0, 'XIVRI': 0.0}}, 'summary': {'raw_runs': {}}
```

Note there are expected to be minor performance differences moving from environment to environment. This include both software and hardware. These difference we notice are usually small, in this case (dev machine vs testbench) all  $\leq 0.1\%$ . On extreme cases this may get bigger, but never big enough to challenge conclusions.



### 3. Train a model

1. limit your card if your psu came from a pawn shop or smth, or if you have an expensive GPU.

su

```
nvidia-smi -pm 1 && nvidia-smi -pl 100
```

2. go to a model dir, say the example playground dbgtrain-all-cntralgn-lcam-ctc-32-128-SASTT

```
cd ~/cat/uni-zero/neko_2025_NGNW/beta320/dbgtrain-all-cntralgn-lcam-ctc-32-128-SASTT
```

3. Train

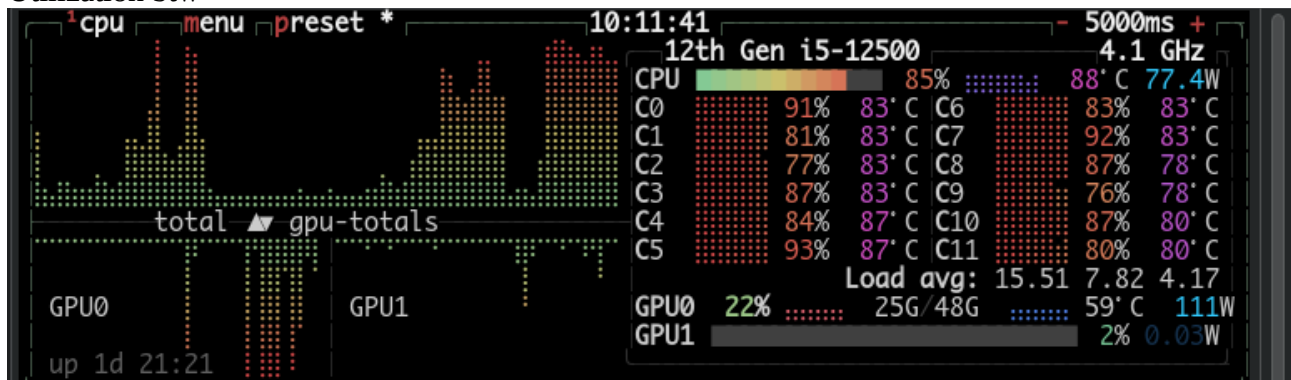
```
PYTHONPATH=${PWD}/../../../../../ python train.py | tee hee.log
```

Or just.. run train.py in pycharm for debugging.

If you see this, you are training.

```
Unable to set tracing for existing threads. Result: 200
INFO: anchor index loaded
INFO: anchor index loaded and secured
INFO: loading anchor index /home/lasercat/ssddata/loader_index/punjabi-default_hvr.pkl
INFO: anchor index loaded
INFO: anchor index loaded and secured
INFO: loading anchor index /home/lasercat/ssddata/loader_index/tamil-default_hvr.pkl
INFO: anchor index loaded
INFO: anchor index loaded and secured
INFO: loading anchor index /home/lasercat/ssddata/loader_index/notbg-default_hvr.pkl
INFO: anchor index loaded
INFO: anchor index loaded and secured
INFO: loading anchor index /home/lasercat/ssddata/loader_index/ostv1-default_hvr.pkl
INFO: anchor index loaded
INFO: anchor index loaded and secured
/run/media/lasercat/writebuffer/uni-zero/neko_sdk/neko_framework_N6/neko_modular_N6.py:202: UserWarning: 'parameters' is an empty generator, no gradient clipping will occur.
nn.utils.clip_grad_norm_(this.model.parameters(), this.max_grad_norm, 2)
6% | 30/60000 [02:44<46:44:16, 2.81s/it]
```

Utilization btw



Note this is run on my dev pc. As the model is too big for the poor 1070.

yes I need to upgrade the testbench potatis but I am broke this month lol.

That's all for now, GLHF

